

The Flux Hinge

IB Physics: one area operation, three different physical meanings”

Working rule: For uniform situations, dividing a total by area gives a quantity per unit area; multiplying by the effective area recovers the total. The geometry is portable, but the physical quantity is not: pressure, irradiance and magnetic flux have different units and meanings.

Name: _____

Class: _____

Companion task to *The Placeholder Force*

1. Pressure: force distributed over a surface

A hydraulic piston has area $A = 2.5 \times 10^{-3} \text{ m}^2$. The fluid exerts a uniform normal force of 750 N on it. Ignore edge effects.

A. Represent the surface interaction

Sketch the piston face, its outward normal and the distributed force exerted by the fluid. Explain why pressure describes a normal surface loading rather than a new kind of force.

B. Join local loading to total force

For a uniform pressure,

$$p = \frac{F_{\perp}}{A}, \quad F_{\perp} = pA.$$

Calculate p . Explain why only the component of force perpendicular to the surface is used. Why is pressure a scalar even though force is a vector?

C. Interrogate the word flux

At molecular level, gas pressure can be interpreted as normal momentum transferred to a wall per unit area per unit time. Explain why this is a momentum flux, while $F_{\perp}/A$ is more precisely called normal stress. What physical model is required to connect the two descriptions?

D. Change the geometry

The same force acts on half the area. Predict the pressure. Then explain why an oblique force must be split into normal and tangential components before pressure can be used. What quantity describes tangential surface loading?

2. Solar irradiance: power crossing a surface

Take the Sun’s luminosity as $L = 3.83 \times 10^{26} \text{ W}$ and Earth’s orbital radius as $r = 1.50 \times 10^{11} \text{ m}$. Assume isotropic emission and no atmospheric absorption.

A. Represent geometric spreading

Sketch concentric spheres around the Sun. Mark the same total luminosity crossing two spheres of different radius. Explain why the power is conserved while the power per unit area falls.

B. Calculate the solar irradiance

Use

$$S = \frac{L}{4\pi r^2}$$

to estimate the irradiance at Earth’s orbit in W m^{-2} . Identify what is being divided by area. Why is this a genuine rate flux rather than merely a surface loading?

C. Include projected area

A panel has area 1.8 m^2 and its normal makes an angle of 35° with the incoming rays. Calculate the incident power using

$$P_{\text{in}} = SA \cos \theta.$$

If its electrical efficiency is 22%, estimate the electrical output. Where does the remaining power go?

D. Interrogate the closure

The inverse-square law follows from conserved luminosity and spherical geometry. List three reasons why irradiance at Earth’s surface differs from the value at the top of the atmosphere. Suggest a transmission or efficiency factor that could represent those losses.

3. Magnetic flux: field through an oriented surface

A coil has $N = 200$ turns, each of area $A = 4.0 \times 10^{-3} \text{ m}^2$. It is placed in a uniform magnetic field of flux density $B = 0.25 \text{ T}$. The magnetic field makes an angle of 60° with the normal to the coil.

A. Represent the field and surface

Sketch the coil, its area normal and the magnetic field direction. Mark the angle used in the flux equation. State the units of magnetic flux density B and magnetic flux Φ_B . Why must the angle be measured from the normal rather than from the plane?

B. Apply the uniform-field relation

For a uniform field,

$$\Phi_B = BA \cos \theta.$$

Calculate the magnetic flux through one turn. Explain separately what is represented by B , by $A \cos \theta$, and by Φ_B .

C. Flux linkage and orientation

Calculate the flux linkage $N\Phi_B$. Predict the flux and flux linkage when the coil is rotated so that (i) its normal is parallel to **B** and (ii) its normal is perpendicular to **B**. At which orientation is the flux greatest?

D. Interrogate the word flux

Magnetic flux is not a flow of matter or energy through the coil. What does $BA \cos \theta$ measure? State the additional law needed before a changing magnetic flux predicts an induced emf.

Synthesis: the area operation is shared; the physical quantity is not

1. Complete the comparison.

Case	Local quantity	Total over surface	Units	Physical closure/source
Pressure				
Solar irradiance				
Magnetic flux				

2. Develop the central claim.

Respond to:

“Per unit area is a geometrical operation, not a physical explanation.”

Use all three cases. Include the terms *surface normal*, *projected area*, *normal stress*, *momentum flux*, *irradiance*, *magnetic flux density*, and *magnetic flux*. Explain why pressure is not literally a flux of force, why solar irradiance is a genuine power flux, and why magnetic flux is an area-and-orientation measure without being a material flow.

Teacher guide and indicative answers

The Flux Hinge

1. Pressure: normal stress and momentum transfer

Representation. Pressure acts normally on every small surface element of a fluid boundary. For a fluid at rest there is no preferred tangential direction, so the stress is isotropic and specified by one scalar p .

Calculation.

$$p = \frac{750}{2.5 \times 10^{-3}} = 3.0 \times 10^5 \text{ Pa.}$$

Only the normal force component enters. Pressure is a scalar magnitude; the surface normal determines the direction of the force exerted on a particular face.

Momentum-flux interpretation. In kinetic theory, molecules reverse their normal momentum component at the wall. Summing momentum change per unit time gives force; dividing by wall area gives pressure. Thus pressure equals a normal momentum flux in that model. The macroscopic phrase *force per area* is normal stress, not a flow of force as a substance.

Geometry. Halving the area at fixed force doubles the pressure to 6.0×10^5 Pa. Tangential loading is shear stress. Pressure alone cannot describe it.

Teaching emphasis. Encourage students to separate three levels:

1. contact force on a finite body;
2. stress as local force per area;
3. microscopic momentum transport producing that stress.

The equations are linked, but the words are not interchangeable.

2. Irradiance: conserved power spread geometrically

Geometric spreading. The same luminosity crosses every sphere centred on an ideal isotropic source. Since sphere area grows as $4\pi r^2$, the irradiance falls as r^{-2} .

Solar value.

$$S = \frac{3.83 \times 10^{26}}{4\pi(1.50 \times 10^{11})^2} \approx 1.35 \times 10^3 \text{ W m}^{-2}.$$

This is power crossing unit area normal to the rays, so it is a true energy-transfer rate flux.

Panel.

$$P_{\text{in}} = SA \cos 35^\circ \approx (1.35 \times 10^3)(1.8)(0.819) \approx 1.99 \text{ kW.}$$

At 22% efficiency,

$$P_{\text{el}} \approx 0.438 \text{ kW.}$$

The remainder is reflected, transmitted or converted mainly to thermal energy before being transferred to the surroundings.

Atmospheric and local closure. Absorption by gases and clouds, scattering, reflection, solar elevation, weather and panel temperature all matter. One may package them into transmission and efficiency factors, for example $P_{\text{el}} = \eta \mathcal{T} SA \cos \theta$.

Teaching emphasis. “Solar constant” denotes irradiance, not total solar power. Students should keep L in watts distinct from S in watts per square metre.

3. Magnetism: flux density, area and orientation

Distinct quantities.

$$B : \text{T} = \text{Wb m}^{-2}, \quad \Phi_B : \text{Wb.}$$

At IB level, B is the magnetic flux density or magnetic field strength used in the flux calculation. Φ_B is the total magnetic flux through the chosen surface.

Flux through one turn.

$$\Phi_B = BA \cos 60^\circ = (0.25)(4.0 \times 10^{-3})(0.5) = 5.0 \times 10^{-4} \text{ Wb.}$$

Here B is the field quantity, while $A \cos \theta$ is the projected area perpendicular to the field.

Flux linkage.

$$N\Phi_B = 200(5.0 \times 10^{-4}) = 1.0 \times 10^{-1} \text{ Wb turn.}$$

If the normal is parallel to $\mathbf{B}$, $\theta = 0^\circ$ and the flux is maximum: $\Phi_B = BA = 1.0 \times 10^{-3}$ Wb. If the normal is perpendicular to $\mathbf{B}$, $\theta = 90^\circ$ and the flux is zero.

Interpretation. Magnetic flux measures how much magnetic field passes through the oriented area in the geometrical sense encoded by $BA \cos \theta$. It is not matter or energy flowing through the loop. Faraday’s law, $\mathcal{E} = -N\Delta\Phi_B/\Delta t$, connects a changing flux to an induced emf.

Synthesis and diagnostic points

Case	Local quantity	Total over surface	Units	Closure/source
Pressure	p or normal stress	$F_\perp = pA$	Pa	contact or kinetic model
Solar	irradiance S	$P = SA \cos \theta$	W m^{-2}	luminosity, geometry, transmission
Magnetic	flux density B	$\Phi_B = BA \cos \theta$	T; Wb	field strength, area and orientation

Indicative synthesis. The same area operation can connect pressure to force, irradiance to power, and magnetic flux density to magnetic flux. It does not make those quantities physically equivalent. Pressure is normal stress and can be interpreted microscopically as momentum flux. Irradiance is energy transferred per unit time per unit area. Magnetic flux depends on field strength, area and orientation. Geometry supplies the effective area; the physical context supplies what is distributed across it.

Common misconceptions to probe:

- using “flux” as though something material must literally flow;
- confusing solar luminosity with irradiance;
- treating pressure as a vector or as total force;
- confusing B with total magnetic flux Φ_B , or measuring the flux angle from the surface rather than from its normal.